

Combinatorial Game Theory

Lecture 5

CSEN 1038 Advanced Data Structures and Algorithms

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Introduction: The 21 Game

<https://21-game-ten.vercel.app/>

The 21 Game

- Rules: players alternate turns; each turn removes 1, 2, or 3 stones from one pile.
- Terminal condition: the player who takes the last stone wins.
- Strategy template: force the opponent to receive a multiple of $(m + 1)$.
- For this game ($m = 3$), target states satisfy $N \equiv 0 \pmod{4}$.
- Foundational idea: classify states as **Winning (N)** or **Losing (P)**.

Formalizing Intuition: Game Trees

Formalizing Intuition: Game Trees

State graph for $N = 10$ (allowed moves: $-1, -3, -4$).



black: remove 1 blue: remove 3 green: remove 4

Red nodes are **P-positions**: $\{0, 2, 7, 9\}$ Blue nodes are **N-positions**.

Winner: First player wins by moving $10 \rightarrow 9$, placing the opponent in a **P-position**.

The Power of Symmetry: Cancellation Strategy

The Power of Symmetry: Cancellation Strategy

- For large state spaces, exhaustive game trees are computationally impractical.
- Symmetry can replace brute-force analysis with an invariant-based strategy.
- **Mirroring/Cancellation:** respond to each opponent move with its symmetric counterpart.
- If symmetry is preserved, control returns to the same strategic class each round.
- Example: **Grid Game** — mirror across the central axis.
- Example: **Bowling Game** — cancel by matching removals in paired lanes.

The Core Concept: Game of Nim

Game of Nim and the Core Proof Idea

- Standard Nim: multiple heaps; each move removes any positive amount from exactly one heap.
- Define the **Nim-sum** as bitwise XOR of all heap sizes.
- **Bouton's Theorem:** a state is Loosing iff its Nim-sum equals 0.
- If Nim-sum is nonzero, a move exists that makes it zero.
- If Nim-sum is zero, every legal move makes it nonzero.
- Interactive board prompt: compute a sample XOR position bridging bitwise operations and strategy.

Applications of Nim

Applications of Nim

- Nim provides a blueprint for analyzing many impartial combinatorial games.
- Goal: transform complex rules into independent components with Nim-like behavior.
- Example: **Staircase Nim** with stones distributed across staircase steps.
- Key reduction: only stones on odd-indexed steps contribute to the effective Nim-sum.
- Even-step moves can be interpreted as transitions that cancel under optimal play.
- After reduction, apply standard XOR strategy exactly as in Nim.

Custom Problem: Nim with Skips

Custom Problem: Nim with Skips

- Problem framework: standard Nim, but each player has a limited number of skips.
- A skip consumes one token and passes the turn without changing heap sizes.
- State now includes both heap configuration and remaining skips for each player.
- Discussion: how do skips alter the characterization of P and N positions?
- Discussion: can a skip be modeled as an auxiliary resource or pseudo-heap?
- Interactive goal: propose a recurrence or invariant for this extended game.

Questions?

- https://cp-algorithms.com/game_theory/sprague-grundy-nim.html
- <https://en.wikipedia.org/wiki/Nim>
- <https://www.youtube.com/watch?v=HAq6Fy1Rgmw>
- M. Albert, R. Nowakowski, D. Wolfe, *Lessons in Play: An Introduction to Combinatorial Game Theory*.
- E. R. Berlekamp, J. H. Conway, R. K. Guy, *Winning Ways for your Mathematical Plays*.